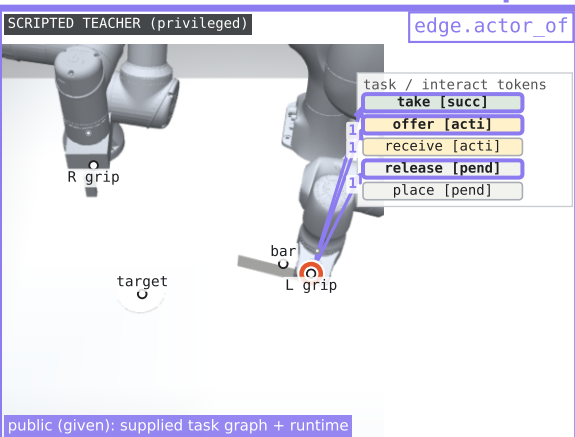
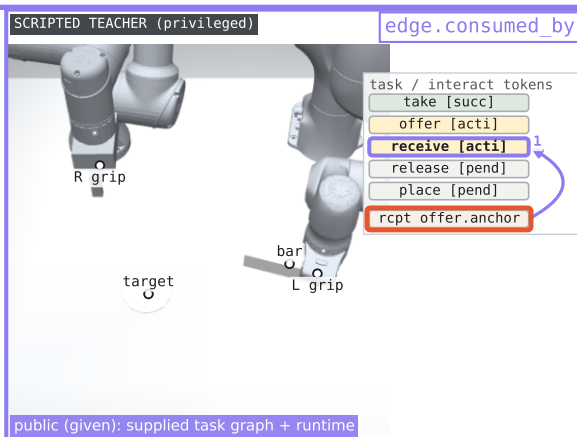


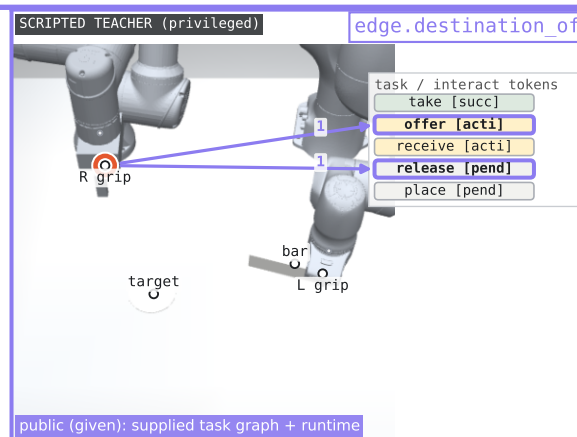
TASK / PROCEDURE (15 implemented)



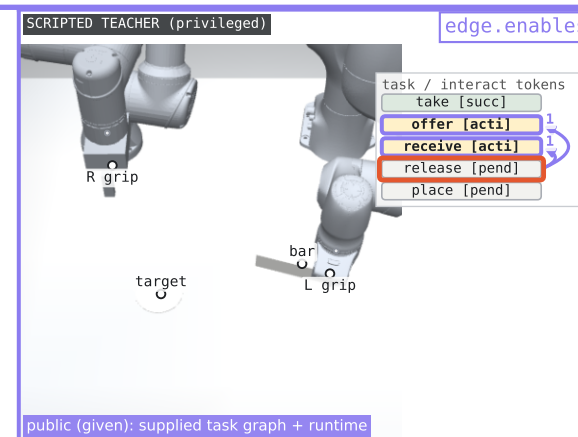
edge.actor_of: manipulator entity ↔ every event it is (cooperating) actor of; L grip acts in take, offer, release. Typed edge channel of the arm-rel-v1 graph, one learned weight per head. edge × bias. Source: public (given): supplied task graph + runtime.



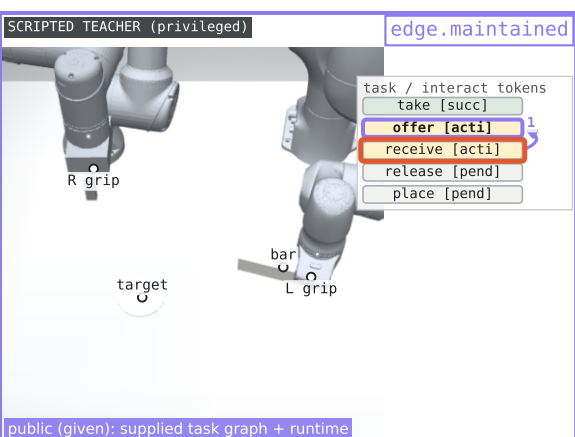
edge.consumed_by: runtime receipt ↔ the event whose output-binding role consumes it: offer.anchor is consumed by receive. edge × bias. Source: public (given): supplied task graph + runtime.



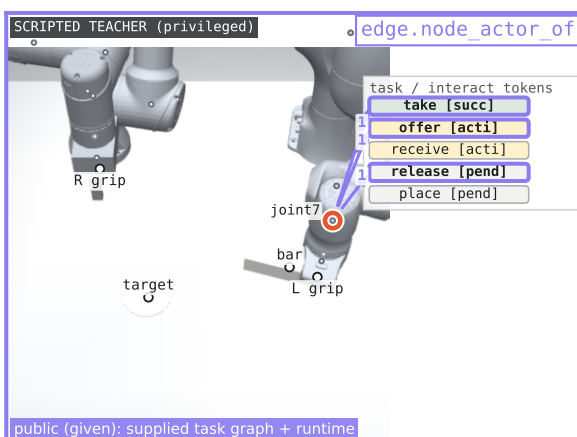
edge.destination_of: destination entity ↔ event: the right gripper is where 'offer' and 'release' deliver the bar (target zone is place's destination). edge × bias. Source: public (given): supplied task graph + runtime.



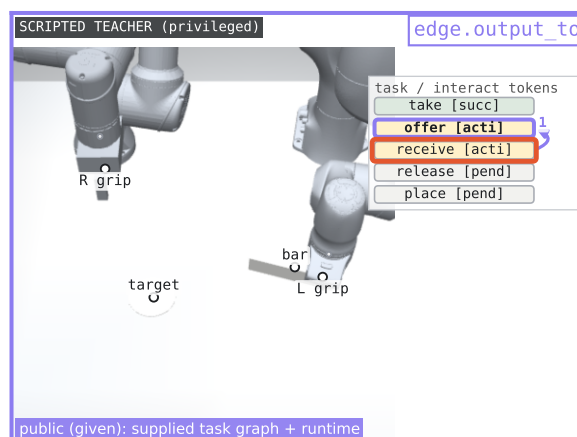
edge.enables: event → event it requires COMPLETED (requires_completed): release needs receive and offer done. Public task-graph precedence. edge × bias. Source: public (given): supplied task graph + runtime.



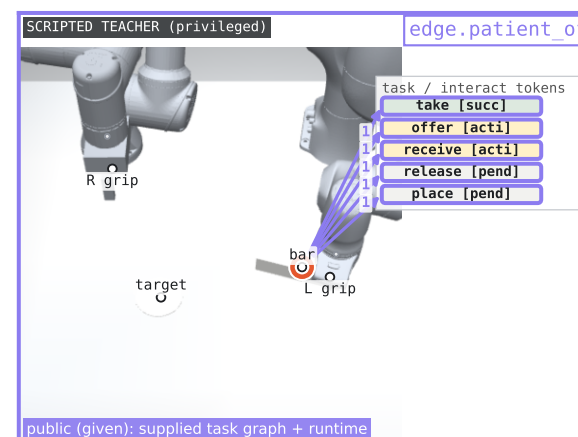
edge.maintained: event → event it requires ACTIVE (requires_active): receive is only valid while offer is held. edge × bias. Source: public (given): supplied task graph + runtime.



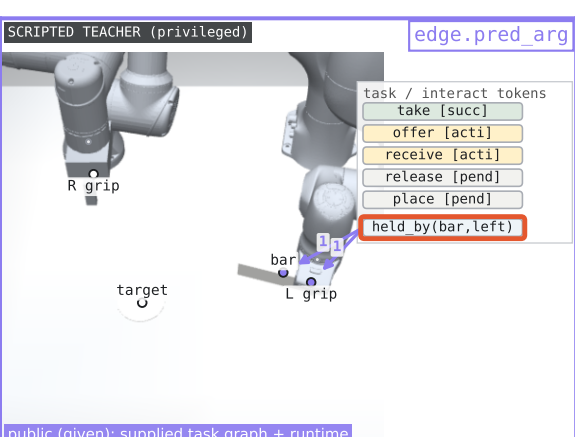
edge.node_actor_of: action node (one joint-command token) → every event its manipulator acts in: a 2-hop composition node → assembly → actor binding. Left-arm joint7 → take, offer, release. edge × bias (act>ctx). Source: public (given): supplied task graph + runtime.



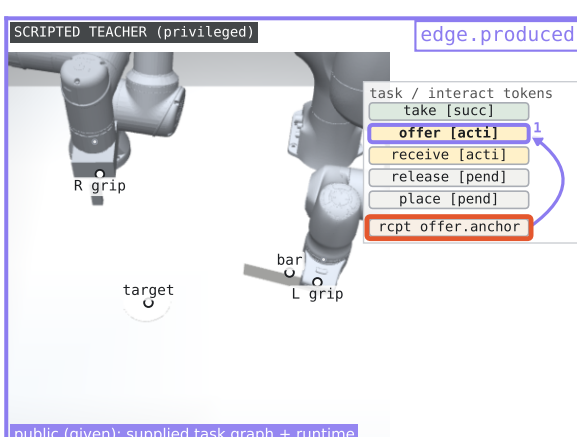
edge.output_to: event ↔ event consuming its output: receive's reference role binds offer's 'anchor' output (event_output binding). edge × bias. Source: public (given): supplied task graph + runtime.



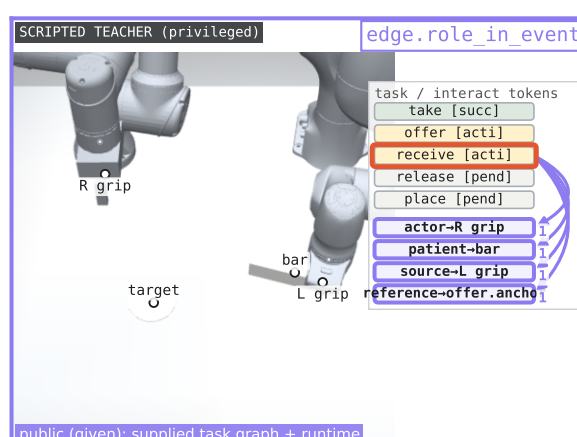
edge.patient_of: patient entity ↔ the events that act on it: the bar is the patient of all five handover events (take, offer, receive, release, place). edge × bias. Source: public (given): supplied task graph + runtime.



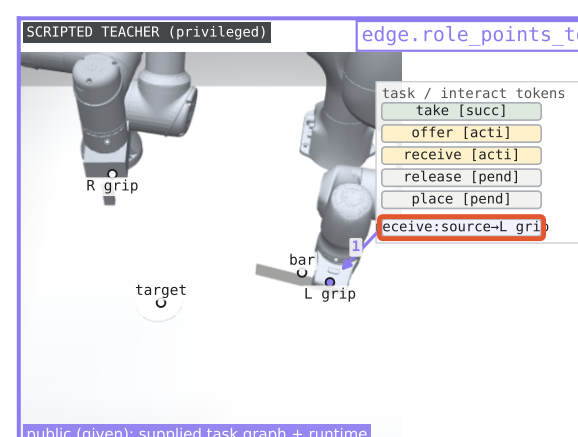
edge.pred_arg: predicate-estimate token ↔ its argument entities: held_by(bar, left) points at the bar slot and the left-gripper assembly. edge × bias. Source: public (given): supplied task graph + runtime.



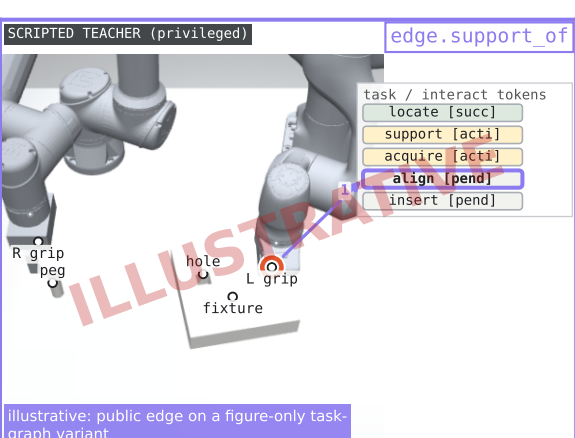
edge.produced: runtime receipt ↔ the event that produced it: the 'anchor' receipt (interact bank) was emitted by offer. edge × bias. Source: public (given): supplied task graph + runtime.



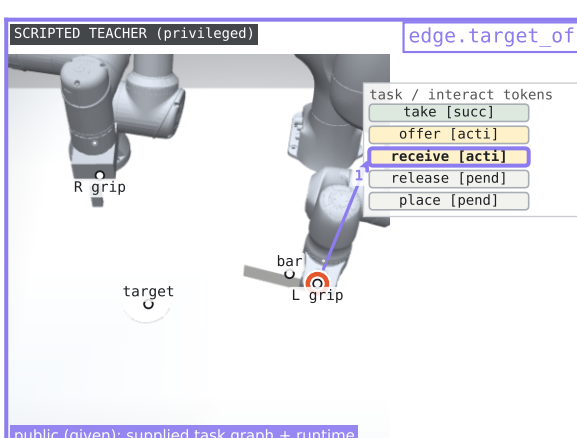
edge.role_in_event: role-slot token ↔ its event: receive owns four ordered role slots (actor, patient, source, reference); multiplicity is explicit. edge × bias. Source: public (given): supplied task graph + runtime.



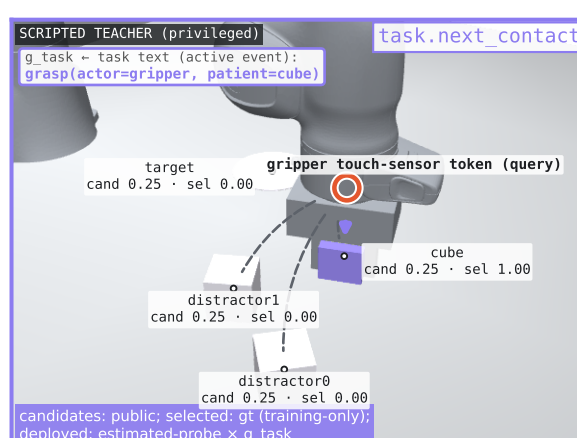
edge.role_points_to: role-slot token → the entity token it is bound to: receive:source → left-gripper assembly (an output binding points to no entity). edge × bias. Source: public (given): supplied task graph + runtime.



edge.support_of: support-role entity ↔ event: the left arm braces the fixture during align. ILLUSTRATIVE: no shipped arm/dual graph binds a support role to an entity; align:support rebound to 'left' (figure-only). edge × bias. Source: illustrative: public edge on a figure-only task-graph variant.

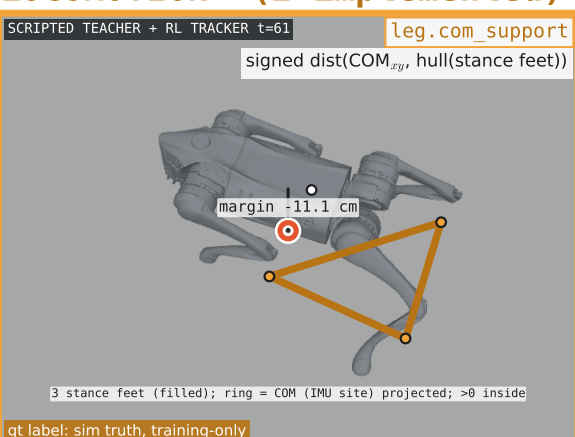


edge.target_of: target entity (and every role without its own channel, here receive:source) ↔ event: the left gripper is the source of 'receive'. edge × bias. Source: public (given): supplied task graph + runtime.

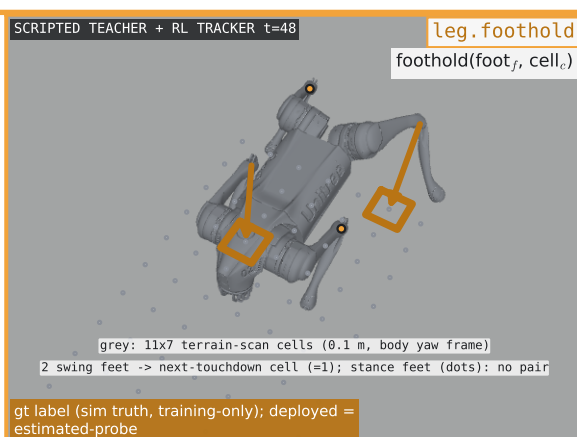


task.next_contact: manipulator → graspable score, sharpened by the task gate (g_task = 1.00 at init, learned). Dashed: PUBLIC candidate edges (1/E|); solid: selected (gt reveal posterior given contact). bilinear × aug, gated. Source: candidates: public; selected: gt (training-only); deployed: estimated-probe × g_task.

LOCOMOTION (2 implemented)



leg.com_support: readout of the stability margin: signed planar distance of the COM projection to the stance-foot support polygon (m, + inside). readout (inert), address assembly, gauss (1-d). Source: gt sim truth (contacts), training-only.



leg.foothold: pair label: each swinging foot → the terrain-scan cell containing its next touchdown (hindsight); planted feet have no true pair. op bilinear, form aug (the bias IS the pair probe). Source: gt label (future touchdown), training-only; deployed = estimated-probe.